Volume 37

Sentinel Distributed Scan Engine (SDSE)

Status: Core Architecture

Priority: Critical

Vision

Scanning should never feel like running a script.

It should feel like launching a mission.

The user simply says:

> Verify the security of all employee laptops.

or

> Assess all production web applications.

Project Sentinel determines how to perform the assessment.

Mission-Oriented Scanning

Instead of:

```
Run Scanner A
```

We use:

```
Mission
```

The scan is only one part of a larger workflow.

Scanner Philosophy

Scanners do not produce findings.

Scanners produce **Evidence**.

Evidence is evaluated later by the Rule Engine.

This separation improves consistency and testability.

Distributed Architecture

```
Mission
```

The planner assigns work efficiently.

Worker Types

Project Sentinel supports multiple worker roles.

Endpoint Worker

Collects evidence from:

- Windows
- Linux
- macOS

Mobile Worker

Collects evidence from:

- Android
- iOS
- Tablets

Within the permissions and platform capabilities provided by the operating system.

Cloud Worker

Collects evidence from:

- AWS
- Azure
- Google Cloud

Through approved APIs and credentials.

Web Worker

Performs assessments of:

- Websites
- APIs

- HTTP services

Only for authorized targets.

Network Worker

Collects:

- Discovery data
- Service metadata
- Configuration evidence

Within authorized network scopes.

Identity Worker

Collects:

- MFA status
- Account configuration
- Role assignments

Container Worker

Collects:

- Container metadata
- Image information
- Runtime configuration

Worker Object

Every worker is itself an object.

```
Worker ID
```

Workers are managed consistently through the Object Kernel.

Scheduling

The scheduler considers:

- Asset priority
- Business hours
- Network load

- Worker health
- Mission deadlines
- Organizational policies

This prevents unnecessary disruption.

Incremental Assessment

After the initial baseline, Project Sentinel avoids collecting unchanged evidence whenever practical.

Benefits:

- Faster assessments
- Lower bandwidth
- Reduced endpoint impact
- Better scalability

Adaptive Planning

If a mission discovers new authorized assets:

```
Mission
```

The mission evolves safely while maintaining an audit trail.

Mobile Device Security

As we discussed earlier, smartphones are now among the most important computing devices.

Project Sentinel should therefore provide dedicated mobile capabilities.

Examples:

- Operating system version verification
- Security patch status
- Device encryption status
- Screen lock configuration
- Installed application inventory (subject to OS permissions)
- Risky permission analysis
- Device integrity indicators available through supported APIs
- Network security configuration
- Backup status
- Certificate inspection
- Enterprise enrollment status

The platform should also help users understand and improve their device security.

Endpoint Security

Project Sentinel should help determine whether an endpoint shows signs of compromise using evidence-based analysis.

Examples include:

- Unexpected persistence mechanisms
- Suspicious startup entries
- Unknown scheduled tasks
- Unusual service configurations
- Security software status
- Driver inventory
- Local firewall configuration
- User account anomalies
- Patch status
- Integrity verification of monitored components

Findings are based on collected evidence and transparent rules rather than unsupported assumptions.

Enterprise Fleet Mode

Organizations can launch one mission against:

```
120,000 Endpoints
```

The architecture is designed to scale horizontally.

Worker Health

Every worker continuously reports:

- CPU usage
- Memory usage
- Queue depth
- Mission count
- Last heartbeat
- Version
- Error rate

This allows the platform to detect unhealthy workers before they affect assessments.

NEW CORE ENGINE

Worker Coordination Engine

Responsibilities include:

- Worker registration
- Load balancing
- Capability matching
- Health monitoring
- Automatic reassignment of failed tasks

NEW CORE ENGINE

Evidence Streaming Engine

Rather than waiting until the end of a mission, workers stream evidence as it becomes available.

Advantages:

- Earlier visibility
- Faster recommendations
- Live dashboards
- Improved resilience if a worker disconnects

NEW CORE ENGINE

Fleet Synchronization Engine

Large organizations often have:

- Offices
- Remote workers
- VPN users
- Traveling employees

This engine coordinates assessments across distributed environments while respecting bandwidth limits and organizational policies.

NEW IDEA

Personal Security Mission

For home users:

One button:

Check My Computer

Project Sentinel performs a guided assessment and presents results in clear language:

- Overall health

- Security improvements
- Privacy recommendations
- Verification status
- Suggested next steps

No technical expertise required.

NEW IDEA

Family Protection Mode

A household can manage:

- Parent devices
- Children’s laptops
- Phones
- Tablets

With consent-based administration, clear privacy controls, and role-based access.

Parents see overall security health without unnecessary exposure to personal content.

NEW IDEA

Enterprise Mission Center

Executives can launch organization-wide assessments.

Examples:

- Verify encryption across all laptops.
- Review MFA adoption.
- Assess production web applications.
- Check certificate expiration.
- Validate backup readiness.

Progress is visualized in real time.

PS-ADR-0032

Evidence Collection Separation

Status: Accepted

Decision

Scanning components shall collect evidence only. Interpretation, policy evaluation, trust calculations, and recommendations shall be performed by their respective engines.

Rationale

Separating evidence collection from decision-making improves transparency, reuse, testing, scalability, and long-term maintainability.

NEW LONG-TERM VISION

Sentinel Security Fabric

I think we can now describe the distributed architecture as a **Security Fabric**.

Instead of isolated scanners, the platform becomes a coordinated network of trusted workers, evidence streams, and mission planners operating under one consistent architecture.

Every assessment—whether it involves a home laptop, an enterprise fleet, or an authorized web application—is executed using the same principles:

- Mission-driven
- Object-centric
- Evidence-first
- Policy-aware
- Explainable
- Auditable

A new idea that aligns perfectly with your original vision

I'd like to introduce the **Sentinel Resource Guardian**.

One common complaint about enterprise security tools is that they consume excessive CPU, memory, battery, or network bandwidth.

Project Sentinel should continuously monitor its own resource usage and adapt automatically.

For example:

- Pause intensive work when a laptop is on battery power.
- Reduce scan intensity during video conferences or high CPU load.
- Resume deferred work when the device is idle.
- Respect maintenance windows defined by the organization.

The goal is for users to feel that Project Sentinel is **always working for them, never getting in their way**.

I think that matches one of your earliest design principles: **powerful security with a simple, unobtrusive user experience**. It also gives us another meaningful differentiator, because organizations are far more likely to deploy a platform that protects their environment without noticeably disrupting everyday work.

=====[USER]=====

Good and continue

=====[ASSISTANT]=====

Excellent. I think we have now completed enough of the architecture that we can start designing the **enterprise-grade capabilities** that distinguish Project Sentinel from both consumer antivirus products and traditional enterprise security suites.

One thing has become very clear to me:

> **Project Sentinel should not be a collection of scanners. It should become the**

Digital Security Operations Center for an organization.

That means it continuously understands the organization's cyber environment.

PROJECT SENTINEL BLUEPRINT

Volume 38